

WARSAW, IN, USA

WMO: 720266

Lat: 41.275N

Lon: 85.840W

Elev: 850

StdP: 14.25

Time Zone: -5.00 (NAE)

Period: 04-19

WBAN: 54809

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	0.3	4.9	-8.2	3.6	1.2	-3.5	4.7	6.3	24.6	25.9	21.9	25.1	6.4	250	0.470

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.6	90.2	73.6	87.9	72.7	84.5	71.1	77.3	86.1	75.4	83.5	73.8	81.6	8.0	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
74.6	133.6	82.4	72.6	124.8	80.2	71.8	121.5	79.2	41.1	85.8	39.5	84.2	38.0	81.7	99.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
20.8	18.4	16.3	-7.8	93.5	6.7	3.7	-12.6	96.1	-16.5	98.3	-20.3	100.4	-25.2	103.1		
			-8.3	79.4	6.9	2.6	-13.3	81.3	-17.3	82.8	-21.1	84.2	-26.1	86.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	50.5	25.4	27.5	38.1	49.6	60.9	69.9	72.8	70.8	65.2	52.8	40.6	31.0	(6)
(7)		DBStd	19.10	11.64	11.23	11.13	9.43	9.01	6.21	5.78	5.27	7.55	9.16	9.28	9.95	(7)
(8)		HDD50	2908	764	634	400	121	14	0	0	1	74	308	592		(8)
(9)		HDD65	6080	1227	1051	835	468	188	23	7	88	393	733	1055		(9)
(10)		CDD50	3094	1	3	33	109	353	598	706	645	456	162	25	3	(10)
(11)		CDD65	789	0	0	2	6	62	170	248	192	94	15	0	0	(11)
(12)		CDH74	6975	0	0	18	61	622	1442	2291	1593	821	126	1	0	(12)
(13)		CDH80	2101	0	0	2	5	173	440	771	424	261	25	0	0	(13)
(14)	Wind	WSAvg	6.3	8.0	7.9	7.7	8.2	6.2	5.1	4.3	3.8	4.4	5.9	6.8	7.3	(14)
(15)	Precipitation	PrecAvg	39.3	2.7	1.9	2.3	3.5	4.4	4.1	4.4	4.4	3.0	3.3	3.0	2.4	(15)
(16)		PrecMax	52.5	5.7	4.6	4.6	7.0	7.0	7.5	7.5	8.4	6.2	8.1	5.3	5.9	(16)
(17)		PrecMin	31.8	0.9	0.6	0.7	0.4	1.6	2.0	2.1	2.3	1.0	0.9	0.8	0.7	(17)
(18)		PrecStd	4.8	1.4	1.1	0.9	1.5	1.8	1.7	1.7	1.7	1.7	2.0	1.0	1.2	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	57.1	61.4	76.6	80.8	89.7	91.1	94.7	90.4	93.3	85.6	70.4	60.6	(19)
(20)		0.4%	MCWB	54.2	53.2	63.2	61.3	70.4	72.4	76.6	74.4	74.2	68.8	56.6	56.3	(20)
(21)		2%	DB	51.9	54.6	67.9	74.9	84.3	88.3	90.1	88.0	86.4	78.7	63.6	54.3	(21)
(22)		2%	MCWB	49.5	50.9	58.3	59.4	67.4	71.6	75.1	73.7	72.4	65.0	55.1	51.5	(22)
(23)		5%	DB	45.8	47.9	61.4	71.7	81.3	85.3	87.7	84.3	82.0	72.4	59.3	48.1	(23)
(24)		5%	MCWB	43.2	44.4	53.1	57.5	65.6	70.6	73.6	72.2	68.8	62.2	52.2	45.3	(24)
(25)		10%	DB	41.0	42.5	55.2	66.3	77.1	81.7	84.1	81.7	79.0	68.2	54.8	44.6	(25)
(26)		10%	MCWB	38.0	38.3	49.3	55.0	64.3	68.7	71.8	70.6	67.3	60.2	48.4	41.6	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	54.7	56.5	63.6	64.7	72.9	77.1	80.2	78.4	81.7	71.9	60.4	56.9	(27)
(28)		0.4%	MCDWB	56.9	59.1	76.1	74.3	83.9	87.0	87.0	85.2	84.1	81.8	65.4	60.2	(28)
(29)		2%	WB	49.5	51.1	59.4	61.9	70.4	74.8	77.6	76.3	74.0	67.8	57.2	51.7	(29)
(30)		2%	MCDWB	51.2	54.3	66.2	70.5	80.2	83.6	86.9	83.7	83.3	74.1	62.2	53.4	(30)
(31)		5%	WB	43.6	44.4	55.1	59.4	68.1	73.2	75.9	74.6	72.0	64.6	53.1	45.6	(31)
(32)		5%	MCDWB	45.5	47.8	60.1	67.5	77.4	81.4	84.5	81.7	79.8	69.6	58.8	47.4	(32)
(33)		10%	WB	38.0	38.2	50.3	56.6	66.2	71.4	74.0	72.8	69.8	61.2	49.5	41.5	(33)
(34)		10%	MCDWB	40.9	41.8	55.7	66.5	74.5	78.6	81.7	79.7	74.8	67.1	55.0	44.4	(34)
(35)	Mean Daily Temperature Range		MDBR	13.3	14.3	16.9	20.3	20.4	19.5	19.6	19.6	20.8	18.5	15.1	11.6	(35)
(36)		5% DB	MCDBR	16.6	20.5	24.1	25.6	24.7	22.7	21.3	22.1	25.1	23.7	21.0	16.8	(36)
(37)			MCWBR	15.0	16.8	16.1	14.2	12.1	9.8	8.9	10.0	12.2	12.7	14.5	14.8	(37)
(38)			MCDBR	16.3	20.0	21.9	22.9	20.8	18.0	18.5	17.4	21.1	19.6	18.8	15.9	(38)
(39)		5% WB	MCWBR	15.5	17.1	15.8	15.7	11.1	9.3	9.6	9.6	12.3	12.9	14.7	14.8	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.295	0.326	0.349	0.400	0.441	0.436	0.442	0.446	0.409	0.352	0.318	0.296	(40)	
(41)		taud	2.401	2.315	2.350	2.219	2.151	2.212	2.200	2.180	2.260	2.421	2.465	2.444	(41)	
(42)		Ebn at Noon	272	279	285	276	266	267	264	259	261	266	262	263	(42)	
(43)		Edh at Noon	25	32	35	43	48	45	45	44	38	29	24	23	(43)	
(44)	All-Sky Solar Radiation	RadAvg	489	758	1120	1471	1687	1911	1928	1692	1389	919	582	410	(44)	
(45)		RadStd	47	98	97	131	157	123	104	93	114	90	80	44	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (52 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page